Inside the economies: channels and global forces

What happens underneath GDP: which demand, budget, money and interest-rate channels synchronize across economies and when, a two-economy case study (the United States and Japan), and the global series from the atlas. The channel anatomy holds the one lead that survives across decades: synchronized short-rate rises two years before global recessions ($p = 0.019$, 18 economies).

Status key

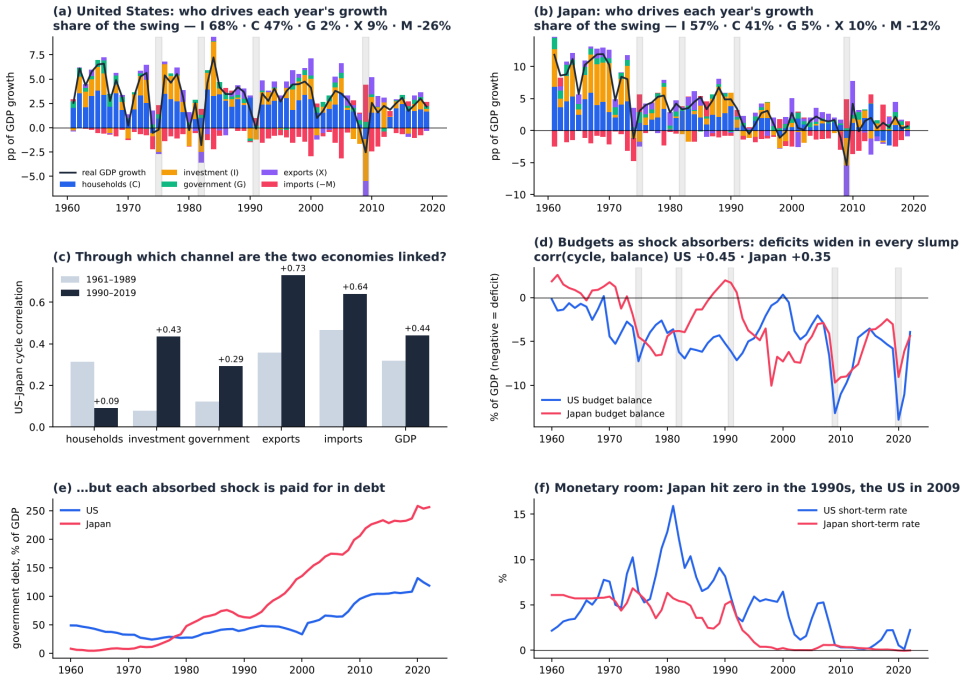
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

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F7.1 DESCRIPTIVE

Inside two economies: the United States and Japan

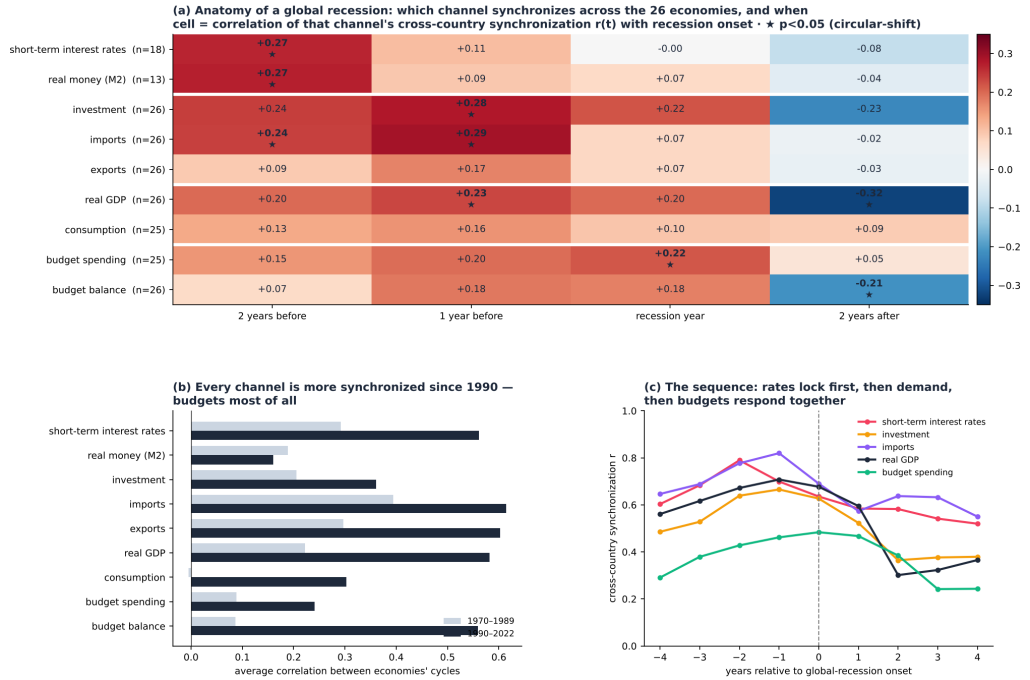


(a, b) Each year's growth split into consumption, investment, government, exports and imports, with each part's share of the variance of growth. (c) US–Japan cycle correlation by channel, 1961–89 and 1990–2019: trade and investment carry the link. (d) Budgets as shock absorbers: deficits widen in every slump. (e) The absorbed shocks accumulate as debt. (f) Monetary room: Japan's short rate reaches zero in the 1990s, the US rate in 2009.

Numbers Variance shares US: I 68%, C 47%, G 2%, X 9%, M −26%; Japan: I 57%, C 41%, G 5%, X 10%, M −12%. Cycle–balance correlation US +0.45, Japan +0.35. Debt 1970→2022: US 33%→119%, Japan 8%→256%.
Made by paper1/scripts/eraB_claude_code/usjp.py (plotting script lost) · **Data** Global Macro Database and Penn World Table 10.01
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F7.2 CURRENT

Anatomy of a global recession: which channel synchronizes, and when

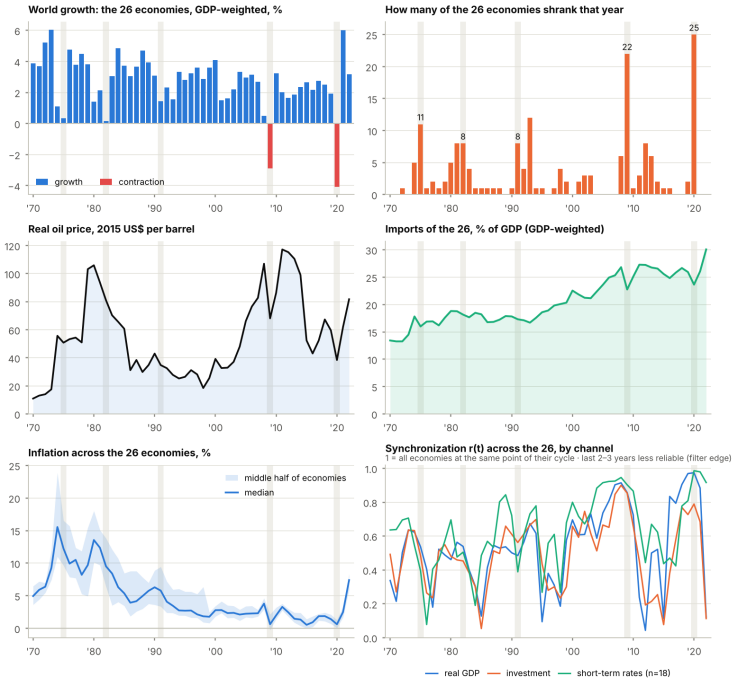


(a) For each channel, the correlation of its cross-country synchronization $r(t)$ with recession onset two years before, one year before, in the recession year and two years after ($\star$ $p < 0.05$, circular shift). Synchronized short-rate rises come two years before; investment and imports one to two years before; budgets in the recession year. (b) Every channel is more synchronized since 1990, budgets most of all. (c) The sequence: rates lock first, then demand, then budgets respond together.

Numbers Short rates (18 economies): +0.27 two years before, p = 0.019.
Made by paper1/scripts/eraB_claude_code/panel26.py → panel26_results.json (reproduces exactly; plotting script lost) · **Data** Global Macro Database
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F7.3 DESCRIPTIVE

Global forces I: growth, oil, trade, inflation and synchronization by channel

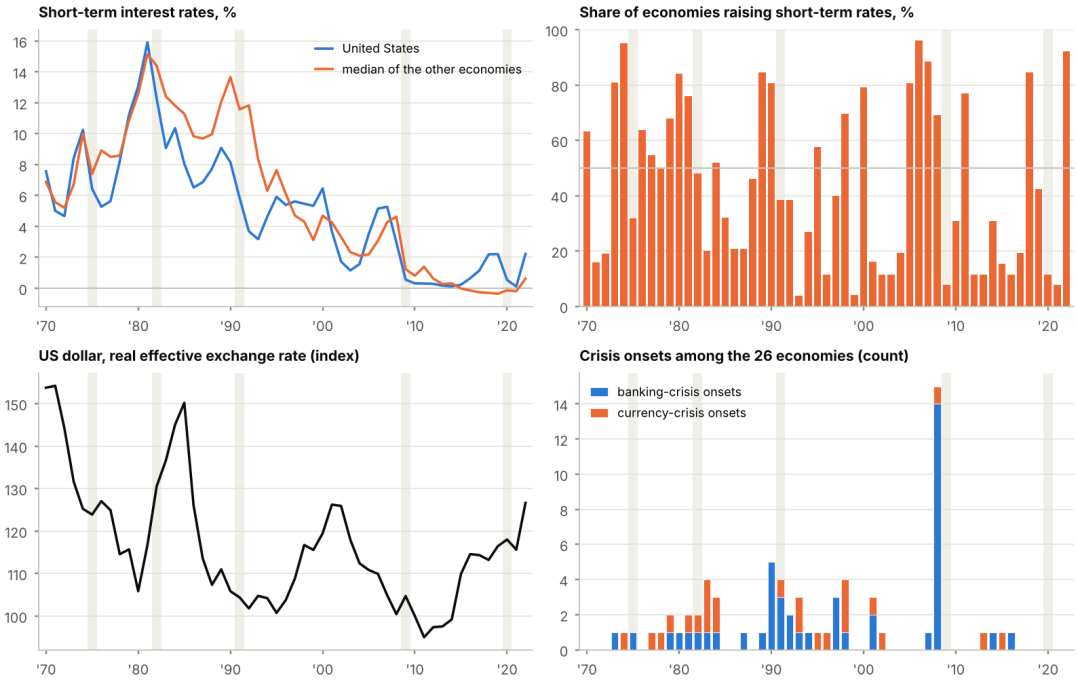


World growth (GDP-weighted), how many of the 26 shrank each year, the real oil price, imports of the 26 as a share of their GDP, inflation across the 26, and $r(t)$ for real GDP, investment and short rates. The last two to three years of each $r(t)$ are less reliable because of the filter edge.

Numbers 25 of 26 shrank in 2020 and 22 in 2009, against 8–11 in 1975, 1982 and 1991. GDP $r(t)$: lead-1 +0.24 (p = 0.019); investment lead-2 +0.29 (0.038); short rates lead-2 +0.27 (0.019).
Made by atlas/figs.py (from atlas/panel.csv, global.csv, stats.json, sync.json) · **Data** Global Macro Database and pinned sources
File figures/07_inside_economies/F7-03_global_real.png

F7.4 DESCRIPTIVE

Global forces II: interest rates, the dollar and crisis onsets

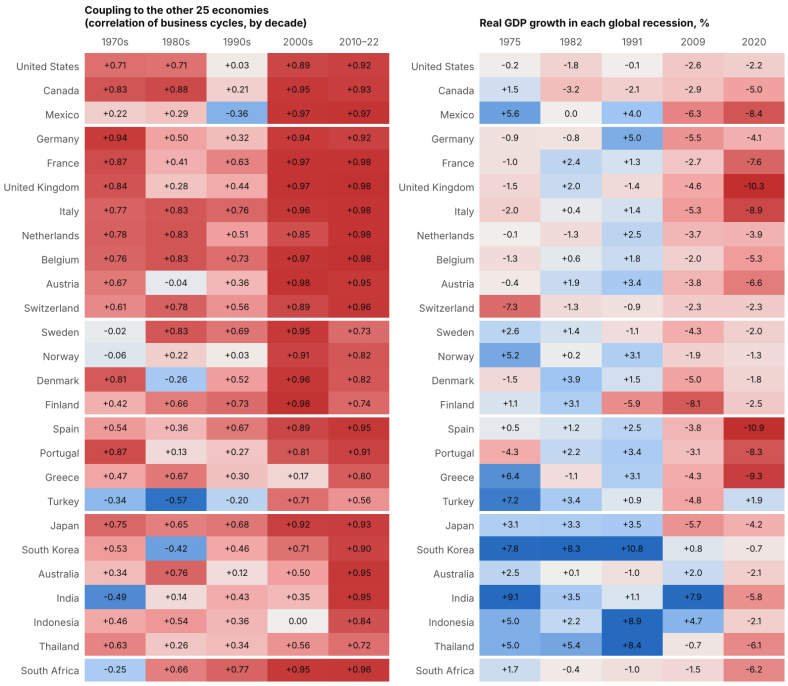


US and median short-term rates, the share of economies raising rates, the US real effective exchange rate, and banking and currency crisis onsets among the 26.

Numbers Descriptive.
Made by atlas/figs.py (from atlas/panel.csv, global.csv, stats.json, sync.json) · **Data** Global Macro Database and pinned sources
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F7.5 DESCRIPTIVE

The 26 side by side: coupling by decade and growth in each recession

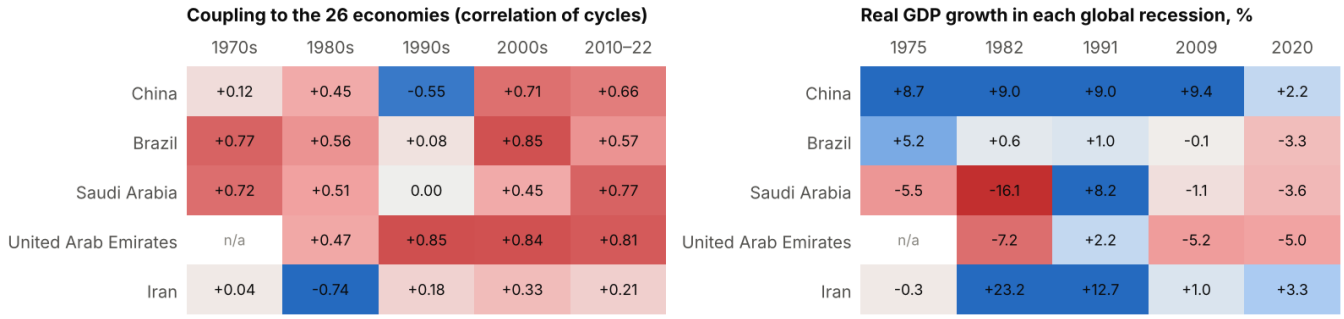


Left: each economy's correlation with the other 25, by decade. Right: real GDP growth in each global recession.

Numbers Average coupling 0.46 (1970–89) and 0.75 (1990–2022).
Made by atlas/figs.py (from atlas/panel.csv, global.csv, stats.json, sync.json) · **Data** Global Macro Database and pinned sources
File figures/07_inside_economies/F7-05_overview_heat.png

F7.6 DESCRIPTIVE

The five added economies: coupling to the 26 and growth in each recession



As the overview heat map, for China, Brazil, Saudi Arabia, the United Arab Emirates and Iran, each compared with the 26 (never part of their averages).

Numbers Descriptive.
Made by atlas/figs.py (from atlas/panel.csv, global.csv, stats.json, sync.json) · **Data** Global Macro Database and pinned sources
File figures/07_inside_economies/F7-06_overview_extra.png